



Lesson 8.6 — Confidence Intervals

Big idea: A confidence interval is a range of plausible values for a population parameter. The level (e.g., 95%) describes how often the procedure produces an interval that catches the true value, IF you repeated it many times.

Quick review

For a proportion: $p \pm z^* \sqrt{p(1-p)/n}$. z^* depends on confidence level ($z^* = 1.96$ for 95%).

For a mean (large n): $\bar{x} \pm z^* (s/\sqrt{n})$.

Interpretation (correct): we are 95% confident the true parameter lies in this interval — meaning the method captures it 95% of the time over many repetitions.

Interpretation (incorrect): the interval has a 95% probability of containing the parameter (the parameter is fixed; the interval is what varies).

Conditions: random sample; for proportions, $np \geq 10$ and $n(1-p) \geq 10$; for means, $n \geq 30$ or population approximately normal.

Worked example

Worked example. 95% confidence interval for proportion: $p = 0.40$, $n = 200$.

$ME = 1.96 \sqrt{(0.4 \cdot 0.6 / 200)} = 1.96 \sqrt{(0.0012)} \approx 1.96(0.0346) \approx 0.068$. CI: $0.40 \pm 0.07 = (0.33, 0.47)$.

Warm-ups

1. What z^* corresponds to 95% confidence?
2. What z^* corresponds to 90% confidence?
3. If a 95% CI is (0.42, 0.48), what is the point estimate?

Practice

1. Compute a 95% CI for p when $p = 0.65$ and $n = 300$.
2. Compute a 90% CI for the mean if $\bar{x} = 50$, $s = 10$, $n = 100$. (Use $z^* = 1.645$.)
3. If a 95% CI for a proportion is (0.30, 0.40), what is ME and p ?

Challenge

1. A survey of 500 students gives $p = 0.62$. Construct a 99% CI ($z^* = 2.576$) and interpret it in plain English.
2. A 95% CI is found to be (62, 70) for the mean. What sample size n would (approximately) halve the width of the CI, with the same s ?



Answer key — Lesson 8.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 1.96.
2. 1.645.
3. $(0.42 + 0.48)/2 = 0.45$.

Practice

1. $ME = 1.96 \sqrt{(0.65 \cdot 0.35 / 300)} \approx 1.96 \sqrt{(0.000758)} \approx 1.96(0.02754) \approx 0.054$. CI: (0.596, 0.704).
2. $ME = 1.645(10/\sqrt{100}) = 1.645$. CI: (48.36, 51.64).
3. $ME = (0.40 - 0.30)/2 = 0.05$. $p \blacksquare = (0.30 + 0.40)/2 = 0.35$.

Challenge

1. $ME = 2.576 \sqrt{(0.62 \cdot 0.38 / 500)} \approx 2.576(0.02171) \approx 0.0559$. CI: (0.564, 0.676). Plain English: we are 99% confident that between 56.4% and 67.6% of the population would respond this way.
2. Width is proportional to $1/\sqrt{n}$. To halve the width (same s, same z^*), n must quadruple.